Jared Siegel

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Education

Princeton University | PhD Candidate in Astrophysics 2022 – Present

Thesis Advisors: Alexandra Amon and Jenny Greene

Research: Observational cosmology and galaxy evolution; weak lensing; baryonic feedback; intrinsic alignment of galaxies

University of Chicago | BA in Physics & BS in Astrophysics 2018 – 2022

Thesis Advisor: Leslie Rogers

Awards and Grants

Harold W. Dodds Fellowship | Princeton University 2026 – 2027

"... recognizes exceptional students in their later years of study ... provides full university fellowship tuition and a 12-month premium stipend."

Graduate Research Fellowship | Jane Street 2026 – 2027

"... supports exceptional doctoral students currently pursuing a PhD in electrical engineering and computer science, mathematics, physics, or statistics."

Graduate Research Fellowship | National Science Foundation 2022 – 2026

Centennial Fellowship | Princeton University 2022 – 2027

Chambliss Astronomy Student Award | American Astronomical Society 2020

Observing Programs

PI James Webb Space Telescope Cycle 4 GO #8607 2025

"Studying cosmic noon at 200 parsec scales: resolved spectroscopy of a magnified dusty quiescent galaxy"

Peer-Reviewed Publications

14 first author; 3 co-author with major contribution; 23 total

Cosmology and Galaxies

9. **Siegel, J.** et al., "Evidence for a thermal pressure deficit in galaxy groups from the tSZ effect and weak lensing", submitted to ApJ, arXiv:2608.00164
8. Shavelle, K., **Siegel, J.**, et al., "A two-component profile of baryonic feedback: hydrostatic and diffuse gas", submitted to MNRAS, arXiv:2609.21861

7. Bigwood, L., McCullough, J., **Siegel, J.**, et al., "Confronting cosmic shear astrophysical uncertainties: DES Year 3 revisited", submitted to PRD, arXiv:2512.04209

6. Bigwood, L., Yamamoto, M., **Siegel, J.**, et al., "The kinetic Sunyaev-Zeldovich effect as a benchmark for AGN feedback models in hydrodynamical simulations: insights from DESI + ACT", MNRAS, 550 (2026) arXiv:2510.15822

5. **Siegel, J.** et al., "The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy lensing", MNRAS, 549, 1 (2026), arXiv:2512.02954

4. **Siegel, J.** et al., "Joint X-ray, kinetic Sunyaev-Zeldovich, and weak lensing measurements: toward a consensus picture of efficient gas expulsion from groups and clusters", ApJ, 1003, 151 (2026), arXiv:2509.10455

3. **Siegel, J.** et al., "Intrinsic alignment demographics for next-generation lensing: Revealing galaxy property trends with DESI Y1 direct measurements", MNRAS, 552, 1 (2026), arXiv:2507.11530

2. **Siegel, J.** & Melchior, P., "Spatially Resolved Galaxy-Dust Modeling with Coupled Data-Driven Priors", ApJ, 986, 212 (2025), arXiv:2411.08111

1. **Siegel, J.** et al., "UNCOVER: Significant Reddening in Cosmic Noon Quiescent Galaxies", ApJ, 985, 125 (2025), arXiv:2409.11457

Extrasolar Planets

5. **Siegel, J.** et al., "Quiet Please: Detrending Radial Velocity Variations from Stellar Activity with a Physically Motivated Spot Model", AJ, 168, 158 (2024), arXiv:2408.07121

4. **Siegel, J.**, Winn, J., & Albrecht, S., "Ponderings on the Possible Preponderance of Perpendicular Planets", ApJ Letters, 950, L2 (2023), arXiv:2305.13400

3. **Siegel, J.** & Rogers, L., "Mass Upper Bounds for Over 50 Kepler Planets Using Low-S/N Transit Timing Variations", AJ, 164, 139 (2022), arXiv:2208.14398

2. **Siegel, J.**, Rubenzahl, R., Halverson, S., & Howard, A., "Into the Depths: a new activity metric for high-precision radial velocity measurements based on line depth variations", AJ, 163, 260 (2022), arXiv:2204.05810

1. **Siegel, J.** & Fabrycky, D., "Resonant Chains of Exoplanets: Libration Centers for Three-body Angles", AJ, 161, 290 (2021), arXiv:2104.14665

High Energy Astrophysics

3. **Siegel, J.** et al., "Investigating the Lower Mass Gap with Low Mass X-ray Binary Population Synthesis", ApJ, 954, 212 (2023), arXiv:2209.06844

2. **Siegel, J.**, Dwarkadas, V. V., Frank, K., & Burrows, D. N., "Can the Fe K-alpha line reliably predict supernova remnant progenitors?", ApJ, 922, 67 (2021), arXiv:2109.01157

1. **Siegel, J.**, Dwarkadas, V. V., Frank, K., & Burrows, D. N., "Analysis of XMM-Newton Observations of Supernova Remnant W49B and Clues to the Progenitor", ApJ, 904, 175 (2020), arXiv:2010.04765

Students Co-Supervised

Kate Shavelle | Princeton Graduate Student — Fall 2025 – Present

Daryna Yushchenko | Princeton Undergraduate Student — Summer 2026 – Present

Selected Talks

Cosmology and galaxy astrophysics with simulations and ML	CCA, Jan. 2027
ITC Seminar	Harvard University, Oct. 2026
Monday Afternoon Seminar [*Invited*]	MIT, Oct. 2026
Tea Talk	Stanford University, Oct. 2026
Cosmology Seminar [*Invited*]	UC Berkeley, Oct. 2026
Cosmology Seminar	Caltech, Sep. 2026
Cosmic Rhythms	Rio de Janeiro, Sep. 2026
COSMO26	Leiden, Aug. 2026
Baryons@Princeton	Princeton University, May 2026
Weekly Seminar [*Invited*]	Harvard University, Apr. 2026
CMB/LSS Seminar	University of Cambridge, Apr. 2026
Physics Seminar [*Invited*]	University of Alabama, Apr. 2026
Cosmology Seminar [*Invited*]	UC Berkeley, Mar. 2026
AGN & Energy Flows Workshop	Center for Computational Astrophysics, Feb. 2026
Mock NYC Workshop	Center for Computational Astrophysics, Jan. 2026
Cosmology Seminar [*Invited*]	Yale University, Nov. 2025
European Southern Observatory MMC2025	Italy, Oct. 2025
Cosmic Ecosystems Conference	Perimeter Institute, Canada, Jul. 2025
Millimeter Universe Conference	University of Chicago, Jun. 2025
Big Data, Big Questions: The Future of Cosmological Surveys	Germany, May 2025

Outreach and Service

Princeton Prison Teaching Initiative — Spring 2025 – Summer 2026

Developed curricula, served as an in-class tutor, and oversaw the teaching team for AST 207 "Gateway to Science" at East Jersey State Prison.

Teaching

Teaching assistant—Princeton University

AST 401 Cosmology	Spring 2026
APC 524 Software Engineering for Scientific Computing	Fall 2025
AST 207 Gateway to Science: Observational Astronomy	Spring 2025
AST 205 Planets in the Universe	Fall 2023

Teaching assistant—University of Chicago

ASTR 211 Computational Techniques in Astrophysics	Spring 2021
ASTR 205 Introduction to Python Programming	Fall 2020